

Yash Raj [25k-0878]

Ex: 1.3

```
1 #include<iostream>
2 using namespace std;
3
4 class car{
5     public:
6         string brand;
7         string model;
8         void Display(){
9             cout<<"Brand: "<<brand<<endl<<"Model: "<<model;
10        }
11 };
12 int main(){
13     car c1;
14     c1.brand = "Suzuki";
15     c1.model = "Alto";
16     c1.Display();
```

Brand: Suzuki
Model: Alto

Task-1

```
1 #include<iostream>
2 using namespace std;
3
4 class teacher{
5     public:
6         string name;
7         string subject;
8         void DisplayInfo(){
9             cout<<"Name: "<<name<<endl<<"Subject: "<<subject<<endl;
10        }
11 };
12 int main(){
13     teacher t1;
14     t1.name = "Javed";
15     t1.subject = "Pakistan Studies";
16     teacher t2;
17     t2.name = "Zameer-Ul-Hassan";
18     t2.subject = "Physics";
19     t1.DisplayInfo();
20     t2.DisplayInfo();
21 }
```

Name: Javed
Subject: Pakistan Studies
Name: Zameer-Ul-Hassan
Subject: Physics

Task – 2

```
1 #include <iostream>
2 using namespace std;
3
4 class BankAccount {
5     private:// Private Members protects object's sensitive data from unauthorized external manipulation.
6         int balance;
7         string AccountNumber;
8     public:
9         void setAccountNumber(string accN){
10             AccountNumber = accN;
11         }
12         void showAccount(){
13             cout<<"Account Number: "<<AccountNumber<<endl;
14         }
15         void setBalance(int b) {
16             balance = b;
17         }
18         void showBalance() {
19             cout << "Balance: " << balance << endl;
20         }
21 };
22
23
24 int main() {
25     BankAccount b1;
26     b1.setAccountNumber("P1204-8889777-098");
27     b1.setBalance(10000);
28     b1.showAccount();
29     b1.showBalance();
30 }
```

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```
Account Number: P1204-8889777-098
Balance: 10000
```

Ex: 3.3

```
1 #include <iostream>
2 using namespace std;
3 class Student{
4     private:
5         float cgpa;
6     public:
7         void setCgpa(float c){
8             if(c > 4.0){
9                 cout<<"Invalid! Must be Under 4.0"<<endl;
10            }else if(c < 0){
11                cout<<"Must be Positive"<<endl;
12            }else{
13                cgpa = c;
14            }
15        }
16        void getCgpa(){
17            cout<<"Cgpa: "<<cgpa;
18        }
19 };
20 int main(){
21     Student s1;
22     s1.setCgpa(3.8);
23     s1.getCgpa();
24 }
```

```
Cgpa: 3.8
```

Task – 4.4

Header File: Book.h

```
1 #ifndef BOOK_H
2 #define BOOK_H
3 #include<string>
4 using namespace std;
5
6 class Book{
7     private:
8         string Title;
9         string Author;
10        float Price;
11    public:
12        void setInfo(string title,string author,float price);
13        void getInfo();
14 };
15 #endif
```

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```
1  #include<iostream>
2  #include "Book.h"
3  using namespace std;
4
5  void Book::setInfo(string title,string author,float price){
6      Title = title;
7      Author = author;
8      Price = price;
9  }
10 void Book::getInfo(){
11     cout<<"Title: "<<Title<<endl;
12     cout<<"Author: "<<Author<<endl;
13     cout<<"Price: "<<Price<<endl;
14 }
15
16 int main(){
17     Book b1;
18     b1.setInfo("Animal Farm","George Orwell",350);
19     b1.getInfo();
20 }
```

```
Title: Animal Farm
Author: George Orwell
Price: 350
```